Currently Working On: Learnable gamma-Gating and Dissipation Ablation

Date: July 10, 2026

We are actively enhancing the **Lindblad Neural Operator (LNO)** framework to transition from a fixed dissipation model to a data-driven, physics-informed **"Glass Box"** architecture. This module introduces learnable gating mechanisms and sparse regularization to explicitly isolate, quantify, and prove the utility of specific Lindblad noise channels.

1. Architectural Evolution: Learnable γ-Gating

To determine whether individual dissipation pathways are physical or merely overparameterized artifacts, we are replacing plain, continuous dissipation parameters with an explicit structural gate using a composite activation function:
$$\gamma_k = \text{Softplus}(\text{raw}\_\gamma) \odot \sigma(\text{raw}\_\gamma)$$

- **$\text{Softplus}(\cdot)$:** Guarantees strict positivity ($\gamma_k \ge 0$), satisfying the mathematical constraints of true quantum dynamical semigroups.
- **$\sigma(\cdot)$ (Sigmoid):** Acts as a structural gate, allowing the network to smoothly drive non-contributing dissipation channels exactly to zero ($\gamma_k \to 0$).

To enforce efficiency, we inject an L_1 **Sparsity Penalty** into the foundational objective function:
$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{physics}} + \lambda \sum_{k=1}^{K} |\gamma_k|$$
(Where $\lambda \in [0.001, 0.1]$ modulates structural pruning).

2. Validation Framework: The 3-Figure Proof

To verify that stochastics and specific noise channels are actively driving the model's accuracy ($\text{RMSE} \approx 0.001233$), we are embedding a validation protocol evaluated across three explicit benchmarks:

Metric A: Deterministic vs. Stochastic Ablation ("Noise ON vs. Noise OFF")

We perform a zero-retraining inference test by temporarily clamping the parameters of the dissipation network ($\text{dissipation}\_\text{net} \to 0$) to measure the exact error degradation:
$$\text{Noise Score} = \frac{E_{\text{deterministic}} - E_{\text{stochastic}}}{E_{\text{stochastic}}}$$

- **Interpretation:** A high Noise Score directly verifies that the Lindblad terms are providing foundational regularization rather than fitting random variance.

Metric B: Structural γ-Heatmaps

Extracting the empirical mean of the gated dissipation vectors across the evaluation batch:
$$\bar{\gamma}_k = \frac{1}{B}\sum_{i=1}^{B} \gamma_{k}^{(i)}$$

Plotting these values as a continuous horizontal bar chart exposes exactly which physical dissipation modes the model prioritized, and which unphysical channels were autonomously pruned.

- **Monitored Channels:** [Dephasing, Decay, Thermal, Ion, Measurement, Boundary, Collision, Background]

Metric C: Long-Horizon Energy Drift Stability
Evaluating system dynamics over $T = 1000$ integration steps. We benchmark the current bounded long-horizon trajectory ($\Delta E < 10^{-6}$) directly against a stripped deterministic operator to visually demonstrate how missing noise terms induce unphysical energy blow-ups over extended durations.

3. Pipeline Implementation Target

Phase	Core Mechanism	Objective / Metric	Status
Phase 1	**Zero-Shot "Fake Ablation"**	Compute immediate Noise Score on pre-trained weights by zeroing $\text{dissipation}_\text{net} $ data buffers.	In Progress
Phase 2	**Targeted γ-Fine tuning**	Freeze operator backbone; optimize $\text{dissipation}\_\text{net}$ with L_1 loss over $10\text{--}15$ epochs for structural convergence.	Scheduled
Phase 3	**Manifold Diagnostic Visualization**	Generate composite figures detailing γ-Bar plots, Ablation differentials, and Energy drift fields.	Scheduled